

ARKO

c.arko.business@gmail.com | Remote, US | [linkedin.com/in/arkchakrabarti](https://www.linkedin.com/in/arkchakrabarti) | github.com/SA-Ark

TECHNICAL SKILLS

AI/ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Model Context Protocol (MCP), Vector Search, Prompt Engineering, AI Evaluation

Languages: TypeScript, Python, Rust, SQL

Stack: React, Next.js, Node.js, FastAPI, PostgreSQL, pgvector, Redis, Stripe, Auth.js, WebSocket, Docker, AWS, Cloudflare, CI/CD

PROFESSIONAL EXPERIENCE

Independent AI & Software Consultant 2024 – Present

Principal Engineer — Production SaaS platforms with integrated AI for startup and enterprise clients

NovaBridge Technologies (B2B SaaS Startup, \$8M ARR) — 6 months

- Architected multi-tenant SaaS platform from ground up with role-based access control, real-time collaboration, and Stripe billing, onboarding 200+ business clients within 3 months of launch
- Engineered AI-powered client health scoring system analyzing engagement across 15 data points, identifying at-risk accounts 30 days before churn and recovering \$420K in annual revenue
- Designed WebSocket notification system achieving <100ms delivery latency across 1,000+ concurrent users, increasing platform daily engagement by 28%

Beacon Learning (EdTech Startup, Seed Stage) — 4 months

- Built adaptive learning platform serving 10,000+ students with AI-generated assessments, improving test scores by 40% over static content through personalized difficulty calibration
- Developed AI question generation pipeline converting source material into 50 calibrated questions in under 30 seconds, reducing content creation time from 2 weeks to 1 day
- Implemented RAG-powered AI tutor handling 5,000+ daily queries with 91% resolution rate, reducing instructor support load by 65%

Greenline Commerce (Online Retailer, \$5M Revenue) — 5 months

- Rescued failing e-commerce platform: improved Lighthouse from 32 to 98, eliminated 14 critical security vulnerabilities (SQL injection, XSS, exposed API keys), reduced page load from 8.2s to 1.1s
- Integrated 6 production AI features post-rescue: semantic search (+35% conversion), RAG chatbot (-60% support tickets), recommendation engine (+25% average order value)

AI Data Specialist — Scale AI 2023 (6 months)

- Evaluated large language models across reasoning, code generation, and instruction-following tasks, directly contributing to quality improvements deployed to millions of end users
- Conducted AI safety red-teaming identifying 40+ critical failure modes including prompt injection vulnerabilities and hallucination patterns in production LLM systems

Full-Stack Software Engineer — Contract 2020 – 2023

- Delivered 8 production web applications across fintech, healthcare, and e-commerce verticals, maintaining 99.9% uptime and serving 500K+ combined monthly active users
- Engineered real-time data pipelines processing 2M+ daily events using WebSocket and SSE, powering live dashboards for trading and analytics clients
- Led migration of legacy monolith to microservices for financial services client, reducing deployment time from 3 days to 45 minutes and eliminating 90% of release-related incidents